

ReactJS HOL - 12

Objective

Develop a React application named ticketbookingapp to demonstrate conditional rendering using Login and Logout functionality.

Theory

Explain Conditional Rendering

Conditional rendering allows React components to display different UI elements depending on a condition or application state.

Define Element Variables

Element variables store JSX elements in variables so that different components can be rendered based on conditions.

Prevent Components from Rendering

A component can prevent rendering by returning null or by rendering alternative content based on conditions.

Prerequisites

- Node.js
- npm
- Visual Studio Code

Implementation Steps

1. Create React project: `npx create-react-app ticketbookingapp`
2. Create LoginButton, LogoutButton, GuestGreeting and UserGreeting components.
3. Use a state variable to track login status.
4. Render Guest or User page conditionally.
5. Display Login/Logout buttons appropriately.
6. Run using `npm start`.

App.js

```
import React,{useState} from 'react';

const LoginButton=({onClick})=><button onClick={onClick}>Login</button>;
const LogoutButton=({onClick})=><button onClick={onClick}>Logout</button>;

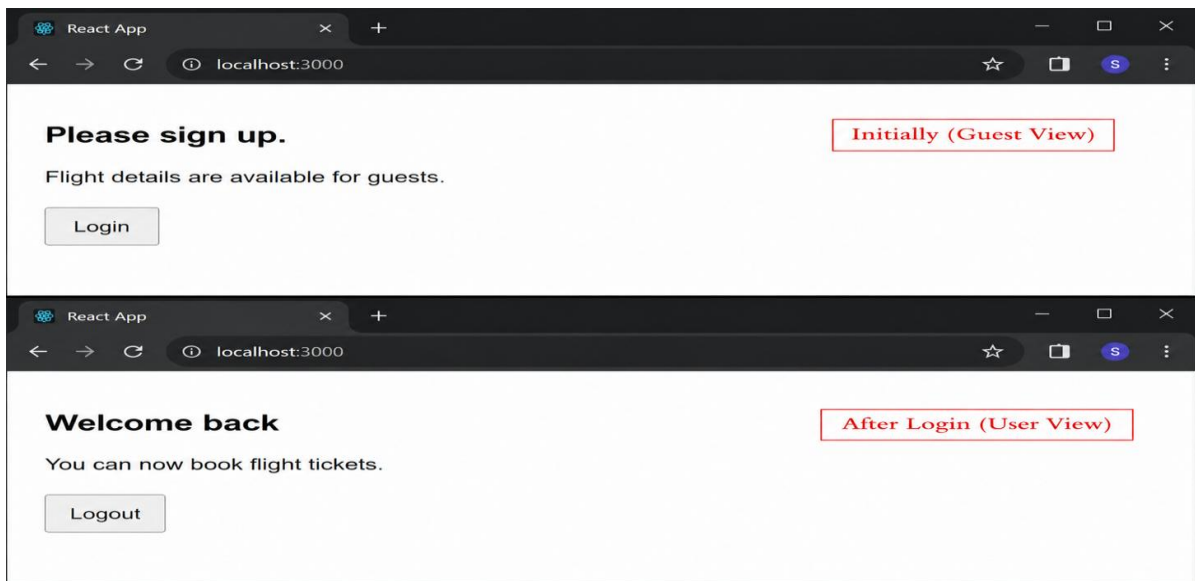
const GuestGreeting=()=>(
<div><h2>Please sign up.</h2><p>Flight details are available for guests.</p></div>);

const UserGreeting=()=>(
<div><h2>Welcome back</h2><p>You can now book flight tickets.</p></div>);

export default function App(){
  const [isLoggedIn,setLoggedIn]=useState(false);
  return(
    <div>
      {isLoggedIn ? <UserGreeting/> : <GuestGreeting/>}
      {isLoggedIn ?
        <LogoutButton onClick={()=>setLoggedIn(false)}> :
        <LoginButton onClick={()=>setLoggedIn(true)}>}
    </div>
  );
}
```

Expected Output

Initially the page displays 'Please sign up.' with a Login button. Clicking Login displays 'Welcome back' with a Logout button. Clicking Logout returns to the guest page.



Learning Outcome

Successfully implemented conditional rendering, element variables, and event handling using React state.

Conclusion

This exercise demonstrated how React dynamically renders components based on login state using conditional rendering.